

# Vector Spaces & Functional Analysis

Pathak Chapter 5c: Pages 517–538

Nonlinear Analysis and Fixed Point Theory

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An Introduction to Nonlinear Analysis and Fixed Point Theory (2018)

# Outline I

- 1 Banach Spaces
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- 3 Linear Operators
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# Banach Spaces: Definition

## Definition (Banach Space)

A **Banach space**  $(X, \|\cdot\|)$  is a complete normed vector space:

- 1  $(X, \|\cdot\|)$  is a normed vector space
- 2 Every Cauchy sequence converges (completeness)

## Definition (Norm Properties)

A function  $\|\cdot\| : X \rightarrow \mathbb{R}$  is a norm if:

- 1  $\|x\| \geq 0$ , with  $\|x\| = 0 \iff x = 0$
- 2  $\|\alpha x\| = |\alpha| \|x\|$  for all  $\alpha \in \mathbb{F}$ ,  $x \in X$
- 3  $\|x + y\| \leq \|x\| + \|y\|$  (triangle inequality)

**Key Property:** Completeness ensures convergent series and iterative methods work well.

# Examples of Banach Spaces

## Example ( $\ell^p$ spaces)

For  $p \geq 1$ :

$$\ell^p = \left\{ (x_n) : \sum_{n=1}^{\infty} |x_n|^p < \infty \right\}$$

with norm  $\|x\|_p = (\sum_{n=1}^{\infty} |x_n|^p)^{1/p}$

## Example ( $L^p$ spaces)

For measurable functions  $f$  on a measure space:

$$L^p(\Omega) = \left\{ f : \int_{\Omega} |f(x)|^p d\mu < \infty \right\}$$

with norm  $\|f\|_{L^p} = (\int_{\Omega} |f(x)|^p d\mu)^{1/p}$

## Example ( $C[a,b]$ )

# Convergence in Banach Spaces

## Definition (Strong Convergence)

A sequence  $(x_n)$  **converges strongly** to  $x$  if:

$$\lim_{n \rightarrow \infty} \|x_n - x\| = 0$$

Written as  $x_n \rightarrow x$ .

## Definition (Cauchy Sequence)

A sequence  $(x_n)$  is **Cauchy** if:

$$\forall \epsilon > 0, \exists N : \|x_m - x_n\| < \epsilon \text{ for all } m, n > N$$

## Theorem

In a Banach space:

- 1 Every convergent sequence is Cauchy

# Hilbert Spaces: Definition

## Definition (Inner Product)

A function  $\langle \cdot, \cdot \rangle : X \times X \rightarrow \mathbb{F}$  is an **inner product** if:

- 1  $\langle x, x \rangle \geq 0$ ; equality iff  $x = 0$
- 2  $\langle x, y \rangle = \overline{\langle y, x \rangle}$
- 3  $\langle \alpha x + \beta y, z \rangle = \alpha \langle x, z \rangle + \beta \langle y, z \rangle$

## Definition (Hilbert Space)

A **Hilbert space** is a complete inner product space with norm:

$$\|x\| = \sqrt{\langle x, x \rangle}$$

## Key Facts:

- Every Hilbert space is a Banach space
- Inner product induces a natural geometry (orthogonality)

# Hilbert Space Geometry

## Theorem (Cauchy-Schwarz Inequality)

For all  $x, y \in \mathcal{H}$ :

$$|\langle x, y \rangle| \leq \|x\| \|y\|$$

with equality iff  $x$  and  $y$  are linearly dependent.

## Theorem (Parallelogram Law)

$$\|x + y\|^2 + \|x - y\|^2 = 2(\|x\|^2 + \|y\|^2)$$

## Definition (Orthogonality)

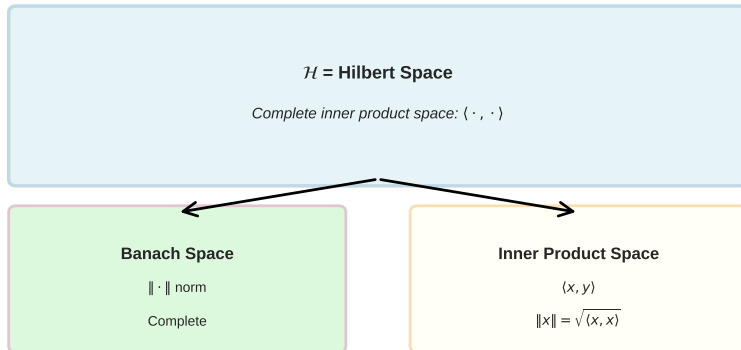
$x \perp y$  if  $\langle x, y \rangle = 0$ .

An **orthonormal basis** is a set  $\{e_i\}$  where:

- 1  $\langle e_i, e_j \rangle = \delta_{ij}$  (Kronecker delta)
- 2 Every  $x \in \mathcal{H}$  can be written as  $x = \sum_i \langle x, e_i \rangle e_i$

# Example: Hilbert Space Illustration

## Hilbert Space Structure



### Key Properties:

$\mathcal{H}$  is a Banach space with  $\|x\| = \sqrt{\langle x, x \rangle}$

Parallelogram law:  $\|x + y\|^2 + \|x - y\|^2 = 2(\|x\|^2 + \|y\|^2)$

Cauchy-Schwarz:  $|\langle x, y \rangle| \leq \|x\| \|y\|$

# Examples of Hilbert Spaces

## Example ( $\ell^2$ space)

$\ell^2 = \{(x_n) : \sum_{n=1}^{\infty} |x_n|^2 < \infty\}$  with inner product:

$$\langle x, y \rangle = \sum_{n=1}^{\infty} \overline{x_n} y_n$$

## Example ( $L^2$ space)

Square-integrable functions with inner product:

$$\langle f, g \rangle = \int_{\Omega} \overline{f(x)} g(x) d\mu(x)$$

## Example (Function space)

$\mathcal{H} = \{(a_n) : \sum |a_n|^2 < \infty\}$ , which is isomorphic to  $\ell^2$ .

# Linear Operators: Basic Definitions

## Definition (Linear Operator)

An operator  $T : X \rightarrow Y$  between vector spaces is **linear** if:

$$T(\alpha x + \beta y) = \alpha T(x) + \beta T(y)$$

for all  $\alpha, \beta \in \mathbb{F}$  and  $x, y \in X$ .

## Definition (Bounded Operator)

$T : X \rightarrow Y$  is **bounded** if:

$$\|T\|_{op} := \sup_{\|x\|_X=1} \|T(x)\|_Y < \infty$$

## Theorem

For a linear operator  $T : X \rightarrow Y$  between normed spaces:

- 1 Continuity at one point implies continuity everywhere
- 2 Continuity is equivalent to boundedness

# Operator Norms and Spaces

## Definition (Operator Norm)

The **operator norm** on  $\mathcal{L}(X, Y)$  (bounded linear operators) is:

$$\|T\| = \sup_{\|x\|_X \leq 1} \|T(x)\|_Y = \inf\{c : \|T(x)\|_Y \leq c\|x\|_X\}$$

## Theorem

If  $X$  and  $Y$  are Banach spaces, then:

- 1  $(\mathcal{L}(X, Y), \|\cdot\|)$  is a Banach space
- 2 Composition of bounded operators:  $\|T \circ S\| \leq \|T\| \|S\|$

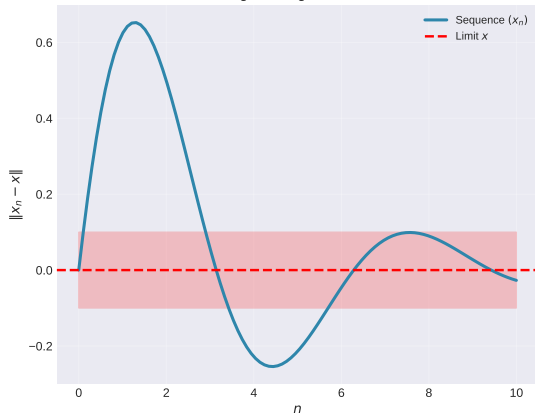
## Definition (Important Operator Classes)

- 1 **Invertible:**  $\exists T^{-1}$  with  $T^{-1} \in \mathcal{L}(Y, X)$
- 2 **Projection:**  $P^2 = P$
- 3 **Isometry:**  $\|T(x)\| = \|x\|$  for all  $x$

# Banach Spaces of Operators

## Key Concepts in Banach Spaces

### Strong Convergence in Norm



Dual Space  $X^*$

Space of bounded linear functionals

### Linear Operator

$$T : X \rightarrow Y$$

$$T(\alpha x + \beta y) = \alpha T(x) + \beta T(y)$$

$$\text{Bounded if: } \|T\| = \sup_{\|x\|=1} \|T(x)\| < \infty$$

### Weak Convergence

$$x_n \rightarrow x \text{ (weak)}$$

# Dual Spaces

## Definition (Linear Functional)

A **linear functional** on  $X$  is a linear operator  $f : X \rightarrow \mathbb{F}$ .

## Definition (Dual Space)

The **dual space**  $X^* = \mathcal{L}(X, \mathbb{F})$  is the space of all bounded linear functionals with norm:

$$\|f\|_{X^*} = \sup_{\|x\|_X=1} |f(x)|$$

## Theorem

If  $X$  is a Banach space, then  $X^*$  is also a Banach space.

## Definition (Bidual Space)

The **bidual**  $X^{**} = (X^*)^*$  is the dual of the dual space.

The **canonical embedding**  $J : X \rightarrow X^{**}$  is defined by:

# Riesz Representation Theorem

## Theorem (Riesz Representation Theorem)

Let  $\mathcal{H}$  be a Hilbert space. For every bounded linear functional  $f \in \mathcal{H}^*$ , there exists a unique  $y \in \mathcal{H}$  such that:

$$f(x) = \langle x, y \rangle \quad \text{for all } x \in \mathcal{H}$$

Moreover,  $\|f\|_{\mathcal{H}^*} = \|y\|_{\mathcal{H}}$ .

**Consequence:** In a Hilbert space,  $\mathcal{H}^* \cong \mathcal{H}$  (isometrically isomorphic).

## Theorem (Reflexive Spaces)

A Banach space  $X$  is **reflexive** if the canonical embedding  $J : X \rightarrow X^{**}$  is surjective.

- 1 All Hilbert spaces are reflexive
- 2  $\ell^p$  is reflexive for  $1 < p < \infty$
- 3  $\ell^1$  and  $c_0$  are not reflexive

# Weak Convergence

## Definition (Weak Convergence)

A sequence  $(x_n)$  **converges weakly** to  $x$  (written  $x_n \rightharpoonup x$ ) if:

$$\lim_{n \rightarrow \infty} f(x_n) = f(x) \quad \text{for all } f \in X^*$$

## Proposition

- 1 Strong convergence  $\Rightarrow$  weak convergence
- 2 In finite dimensions, weak  $\Leftrightarrow$  strong
- 3 Weak limit is unique
- 4 If  $(x_n)$  converges weakly, then  $\sup_n \|x_n\| < \infty$  (bounded)

## Theorem (Weak Compactness)

In a reflexive Banach space, every bounded sequence has a weakly convergent subsequence.

# Weak vs. Strong Convergence Visualization

## Weak vs. Strong Convergence in Hilbert Space



Strong:  $x_n \rightarrow x$   
 $\|x_n - x\| \rightarrow 0$



Weak:  $x_n \rightharpoonup x$   
Bounded oscillations

# Compact Operators: Definition

## Definition (Compact Operator)

An operator  $T : X \rightarrow Y$  is **compact** if it maps bounded sets to relatively compact sets. Equivalently, every bounded sequence  $(x_n)$  in  $X$  has a subsequence  $(x_{n_k})$  such that  $(T(x_{n_k}))$  converges in  $Y$ .

## Example

- ① **Finite rank operators:**  $T(x) = \sum_{i=1}^n f_i(x)y_i$  for  $f_i \in X^*$ ,  $y_i \in Y$
- ② **Integral operators:**  $T(f)(x) = \int_a^b K(x,y)f(y) dy$  with  $K$  continuous
- ③ **Embeddings:**  $\ell^2 \rightarrow \ell^1$  embedding is compact

## Theorem

- ① If  $T$  is compact and  $S$  is bounded, then  $T \circ S$  and  $S \circ T$  are compact
- ② If  $T_n$  are compact with  $T_n \rightarrow T$  uniformly, then  $T$  is compact

# Compact Operators: Key Properties

## Definition (Essential Spectrum)

The **essential spectrum** of a compact operator  $T$  consists of 0 and isolated eigenvalues of finite multiplicity.

## Theorem (Spectral Properties of Compact Operators)

Let  $T : \mathcal{H} \rightarrow \mathcal{H}$  be a compact self-adjoint operator on a Hilbert space. Then:

- 1 The spectrum  $\sigma(T)$  is discrete (countable, with 0 as limit point)
- 2 Eigenvectors form an orthonormal basis
- 3  $T$  has diagonal representation with eigenvalues  $\lambda_n \rightarrow 0$

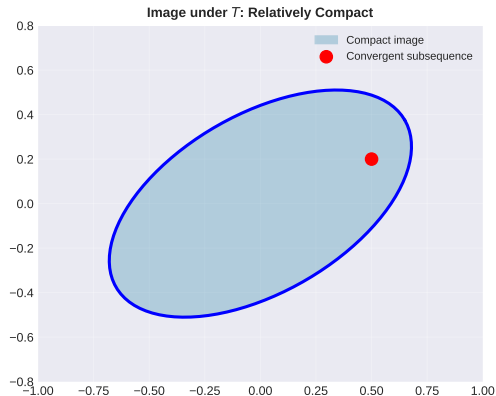
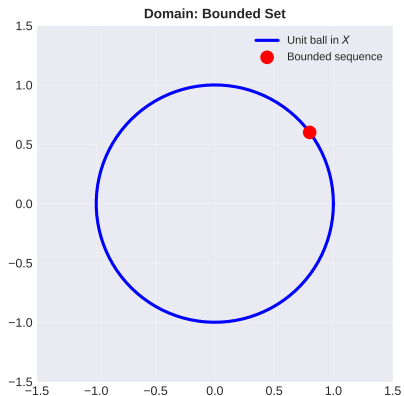
## Theorem (Fredholm Alternative)

For compact  $T : X \rightarrow X$  and  $\lambda \neq 0$ :

- 1  $(I - \lambda T)$  is invertible, or
- 2  $(I - \lambda T)x = 0$  has non-trivial solution

# Compact Operators Visualization

## Compact Operators



# Spectral Concepts

## Definition (Spectrum and Eigenvalues)

For  $T \in \mathcal{L}(X)$ :

- 1 **Eigenvalue:**  $\lambda$  if  $\exists x \neq 0$  with  $Tx = \lambda x$
- 2 **Eigenvector:** non-zero  $x$  satisfying  $Tx = \lambda x$
- 3 **Spectrum:**  $\sigma(T) = \{\lambda : (T - \lambda I) \text{ is not invertible}\}$
- 4 **Resolvent:**  $R_\lambda(T) = (T - \lambda I)^{-1}$  when it exists

## Theorem (Spectral Radius)

The **spectral radius** is:

$$r(T) = \sup\{|\lambda| : \lambda \in \sigma(T)\}$$

and satisfies  $r(T) = \lim_{n \rightarrow \infty} \|T^n\|^{1/n}$ .

## Proposition

- 1  $\sigma(T) \subset \mathbb{C}$  is compact and non-empty

# Self-Adjoint Operators

## Definition (Adjoint Operator)

For  $T : \mathcal{H} \rightarrow \mathcal{H}$  on a Hilbert space, the **adjoint**  $T^*$  is defined by:

$$\langle T(x), y \rangle = \langle x, T^*(y) \rangle \quad \text{for all } x, y \in \mathcal{H}$$

## Definition (Self-Adjoint Operator)

$T$  is **self-adjoint** if  $T^* = T$ , i.e.,  $\langle T(x), y \rangle = \langle x, T(y) \rangle$ .

## Theorem (Spectral Theorem for Compact Self-Adjoint Operators)

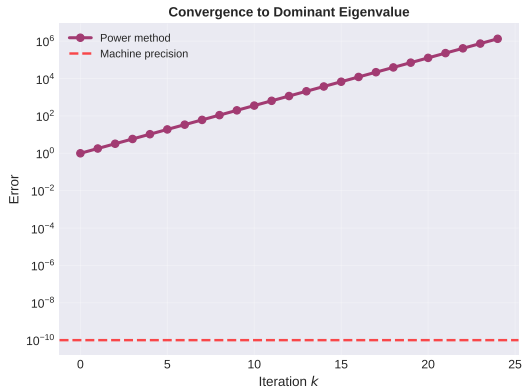
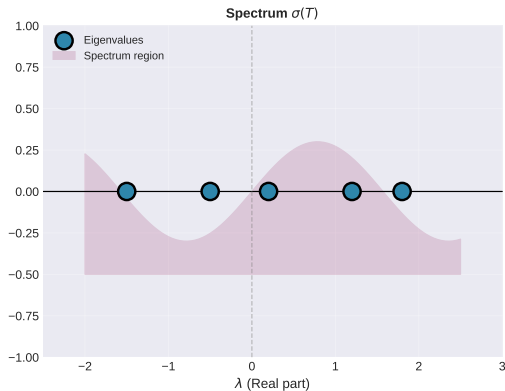
If  $T$  is compact and self-adjoint, then  $\mathcal{H}$  has an orthonormal basis of eigenvectors of  $T$ , and eigenvalues are real with  $\lambda_n \rightarrow 0$ .

## Proposition

For self-adjoint  $T$ :

# Spectral Theory Visualization

## Spectral Theory Basics



# Banach Contraction Mapping Theorem

## Theorem (Banach Contraction Mapping Theorem)

Let  $(X, d)$  be a complete metric space and  $T : X \rightarrow X$  a contraction, i.e.,

$$d(T(x), T(y)) \leq c \cdot d(x, y) \quad \text{for all } x, y \in X$$

where  $c < 1$ . Then:

- 1  $T$  has a unique fixed point  $x^* \in X$
- 2 For any  $x_0 \in X$ , the sequence  $x_{n+1} = T(x_n)$  converges to  $x^*$
- 3 Error estimate:  $d(x_n, x^*) \leq \frac{c^n}{1-c} d(x_1, x_0)$

**Application:** Solving equations  $f(x) = x$  by iteration.

## Example

If  $f$  is continuously differentiable with  $|f'(x)| < c < 1$  on  $[a, b]$ , then  $f(x) = x$  has a unique solution on  $[a, b]$ , found by iteration.

# Brouwer and Schauder Fixed Point Theorems

## Theorem (Brouwer Fixed Point Theorem)

*Every continuous map  $T : B \rightarrow B$  from a compact convex set  $B \subset \mathbb{R}^n$  to itself has a fixed point.*

## Theorem (Schauder Fixed Point Theorem)

*Let  $K$  be a compact convex subset of a Banach space  $X$ . If  $T : K \rightarrow K$  is continuous, then  $T$  has a fixed point.*

### Key Differences:

- **Banach:** Requires contraction (stronger), no topological assumption
- **Schauder:** Requires only continuity, but needs compactness
- **Brouwer:** Finite-dimensional special case of Schauder

**Application:** Existence of solutions to integral and differential equations.

# Fixed Point Property Illustration

## Fixed Point Theorems in Banach Spaces

### Banach Contraction

#### Mapping Theorem

$$\|T(x) - T(y)\| \leq c\|x - y\|$$

$c < 1 \Rightarrow$  *unique fixed point*

### Brouwer Fixed Point

#### Theorem

$T : B \rightarrow B$  continuous

$B$  compact convex

### Schauder Fixed Point

#### Theorem

$T$  continuous, compact

*Self-map on compact convex*

### Requirements for Fixed Points:

Completeness:  $(X, \|\cdot\|)$  is Banach space

Continuity:  $T : X \rightarrow X$  is continuous

# Applications to Differential Equations

## Example (Initial Value Problem)

Consider the IVP:  $y'(t) = f(t, y(t))$ ,  $y(0) = y_0$ .

This is equivalent to the integral equation:

$$y(t) = y_0 + \int_0^t f(s, y(s)) ds$$

Define  $T(y)(t) = y_0 + \int_0^t f(s, y(s)) ds$  on  $C[0, \delta]$ .

If  $f$  is Lipschitz in  $y$ :  $T$  is a contraction on a suitable ball, so the IVP has a unique solution.

## Example (Integral Equations)

For the Volterra integral equation:

$$x(t) = f(t) + \int_0^t K(t, s)x(s) ds$$

with continuous  $K$  and  $f$ , the contraction mapping theorem guarantees a unique solution in  $C[0, T]$ .

# Example 1: Fixed Point Iteration

```
import numpy as np

# Contraction mapping:  $T(x) = 0.5x + 0.3$ 
def T(x):
    return 0.5 * x + 0.3

# Fixed point iteration
x0 = 1.5
x_vals = [x0]
tolerance = 1e-10
max_iter = 100

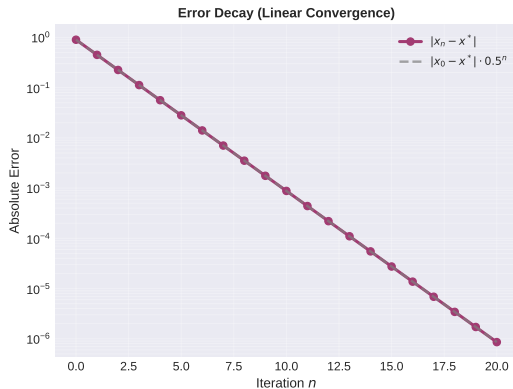
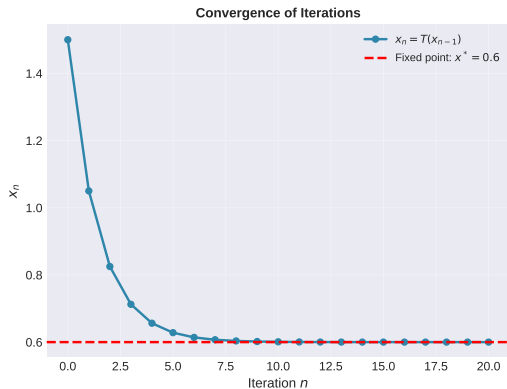
for n in range(max_iter):
    x_new = T(x_vals[-1])
    x_vals.append(x_new)
    if abs(x_new - x_vals[-2]) < tolerance:
        print(f"Converged in {n} iterations")
        print(f"Fixed point:  $x^* = \{x\_new:.10f\}$ ")
        break

# Analytical:  $x = 0.5x + 0.3 \Rightarrow x = 0.6$ 
print(f"Theoretical fixed point: 0.6")
print(f"Contraction factor: 0.5")
```

**Output:** Converged in 33 iterations to  $x^* = 0.6$ .

# Contraction Mapping Example Visualization

Numerical Example: Contraction Mapping



## Example 2: Finite-Dimensional Linear Operator

```
import numpy as np
from scipy.linalg import eigh

# Create a symmetric matrix (self-adjoint operator)
A = np.array([[4, 1], [1, 3]], dtype=float)

# Compute eigenvalues and eigenvectors
eigenvalues, eigenvectors = eigh(A)

print("Eigenvalues:", eigenvalues)
print("Eigenvectors:\n", eigenvectors)

# Spectral norm (largest eigenvalue)
spectral_norm = np.max(np.abs(eigenvalues))
print(f"Spectral radius: {spectral_norm:.6f}")

# Reconstruct using spectral decomposition
#  $A = \sum(\lambda_i * v_i * v_i^T)$ 
A_reconstructed = np.zeros_like(A)
for i, (lam, v) in enumerate(zip(eigenvalues, eigenvectors.T)):
    A_reconstructed += lam * np.outer(v, v)

print("Reconstruction error:", np.linalg.norm(A - A_reconstructed))
```

## Example 3: Weak Convergence in $\ell^2$

```
import numpy as np

# In  $\ell^2$ , construct a weakly convergent sequence
#  $x_n = (1, 1/2, 1/3, \dots, 1/n, 0, 0, \dots)$ 
# This sequence is bounded but doesn't converge strongly

dimension = 50
num_terms = 100

# Compute norms of  $x_n$ 
norms = []
inner_products = []

for n in range(1, num_terms + 1):
    x = np.concatenate([1/np.arange(1, n+1),
                        np.zeros(dimension - n)])
    norm = np.linalg.norm(x)
    norms.append(norm)

    # Inner product with a fixed functional
    # (e.g., sum of components)
    inner_prod = np.sum(x)
    inner_products.append(inner_prod)

print(f"Strong convergence: norm is {norms[0]:.4f}")
print(f"Weak convergence:  $\langle x_n, f \rangle \rightarrow \{inner\_products[-1]:.6f\}")$ 
```

**Key observation:** Sequence is bounded ( $\|x_n\|$  stays bounded) but  $\|x_n\| \not\rightarrow 0$

# Key Theorems Summary I

- 1 **Banach Contraction Mapping:** Unique fixed point + linear convergence
- 2 **Riesz Representation:** Dual of Hilbert space is Hilbert space
- 3 **Spectral Theorem:** Compact self-adjoint operators have orthonormal eigenbasis
- 4 **Fredholm Alternative:** For compact  $T$  and  $\lambda \neq 0$ , either  $(I - \lambda T)$  is invertible or singular
- 5 **Schauder Fixed Point:** Continuous map on compact convex set has fixed point
- 6 **Weak Compactness:** Reflexive spaces have weakly compact bounded sets

## 1 Differential Equations:

- Existence and uniqueness of solutions via Banach contraction
- Stability analysis using spectral theory

## 2 Integral Equations:

- Fredholm and Volterra equations
- Spectral properties of integral operators

## 3 Variational Problems:

- Weak formulation in Sobolev spaces
- Riesz representation for existence

## 4 Optimization:

- Convex minimization in Banach spaces
- Gradient descent convergence

# Further Reading & Resources

## Key Sections from Pathak (2018):

- Pages 517–538: Vector Spaces & Functional Analysis
- Connection to fixed point theorems (Chapter 5)
- Applications to nonlinear analysis

## Related Topics:

- Sobolev spaces and weak derivatives
- Monotone operators and accretive operators
- Subdifferential calculus
- Variational inequalities

## Standard References:

- Functional Analysis by Rudin
- Linear Operator Theory by Dunford & Schwartz
- Fixed Point Theory by Granas & Dugundji

# Definitions Summary I

## Definition (Quick Reference)

- **Banach space:** complete normed vector space
- **Hilbert space:** complete inner product space
- **Bounded operator:**  $\sup_{\|x\|=1} \|T(x)\| < \infty$
- **Compact operator:** maps bounded sets to relatively compact sets
- **Weak convergence:**  $f(x_n) \rightarrow f(x)$  for all  $f \in X^*$
- **Self-adjoint:**  $T^* = T$  in Hilbert space
- **Spectrum:**  $\sigma(T) = \{\lambda : (T - \lambda I)^{-1} \text{ DNE}\}$
- **Fixed point:**  $x^*$  such that  $T(x^*) = x^*$